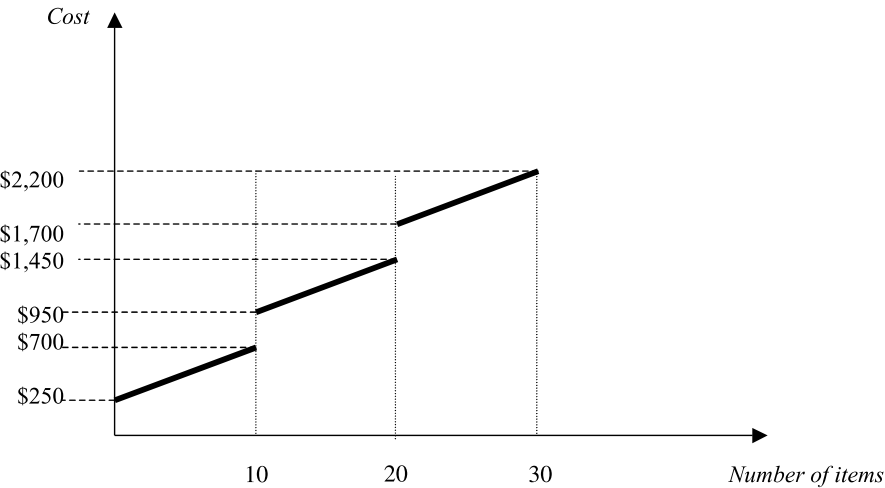


Because of this two-step process, we must realize that we are only finding a solution to the *model* of a problem. If the model is accurate, then the solution will be meaningful. But if the model has too many vague assumptions and approximations, the solution may be meaningless, or worse.

Consider the following example: Suppose a company has 80 warehouses and 5 distribution centers, and every possible route between each warehouse and distribution center has a measurable transportation cost. The shape of this cost function depends on a variety of factors, including the distance between the warehouse and the distribution center, the quality of the road, the traffic density, the number of required stops, the average speed limit, and so forth. The transportation model between warehouse 22 and distribution center 4 might be:

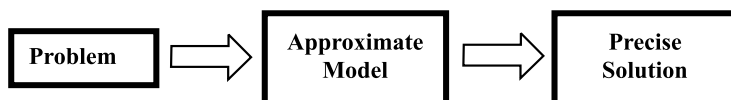


In this model, the cost is zero when there is no delivery. If up to 10 items are delivered, then we incur a fixed cost of \$250 and an additional cost of \$50 per item shipped (thus, the cost for shipping six items would be $\$250 + (6 \times \$50) = \$550$). However, if we transport 11 or more items (but not more than 20), we have to use two trucks. In this case, the cost is \$700 for 10 items, an additional \$250 for the second truck, and \$50 per each additional item (of course, the real situation could be more complex than this, with trucks having different capacities, timetables, etc.). Given the above assumptions, we can construct a model of the entire problem where we specify:

- *All the problem variables.* In this case, there are 400 variables (80 warehouses $\times$ 5 distribution centers), with each variable indicating the number of items to be sent from any warehouse to any distribution center.
- *The constraints that define a feasible solution.* These could be (1) no transport from any warehouse can exceed the number of available items in that warehouse, and (2) the total shipment to any distribution center must satisfy the demand (i. e., the total shipment is at least equal to the number of ordered items).
- *The objective.* In this particular example, we might be concerned with minimizing the total transportation cost.

If our model for transporting items between warehouse 22 and distribution center 4 accurately describes the real-world situation, then we can construct similar models for the other warehouses and distribution centers. But such precise models may prove to be of limited utility. Why? For starters, the cost function in this particular model is discontinuous, and discontinuities present severe difficulties for traditional optimization techniques. Hence, the results that we would obtain from using traditional (e. g., gradient-based⁴) techniques on these functions are likely to be quite poor. And if we cannot derive a solution based on this model, the model – as perfect as it may be – is useless for deciding what to do!

So what options do we have? Well, we can try to simplify the model so that traditional optimization techniques produce better solutions, or we can keep the model unchanged and use a non-traditional approach for finding a near-optimum solution. Put a different way, the first approach uses an approximate model of a problem, and then finds the precise solution for this approximate model:

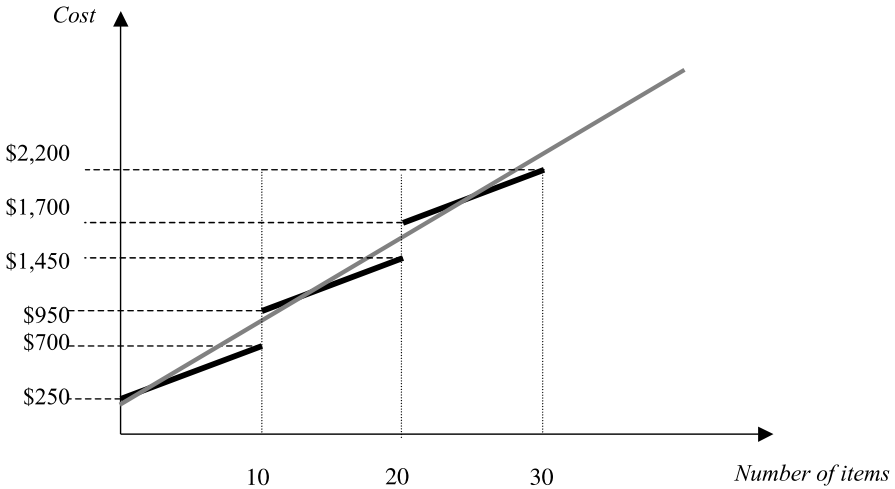


and the second approach uses a precise model of the problem, and then finds an approximate solution for this precise model:



The first approach is quite tempting. For example, we can *approximate* the transportation model between warehouse 22 and distribution center 4 as follows:

⁴ *Gradient* is a rate of change with respect to distance of a variable quantity.



We can simplify the other models in a similar manner, and by making all the models linear we can then use a linear programming method⁵ to find a precise solution. However, this precise solution would be a solution for the simplified model, and not for the real problem!

The second approach is to leave the precise model unchanged – with all its discontinuities and irregularities – and use a non-traditional method to find a near-optimal solution. Of these two approaches, the latter one is often superior (i. e., an approximate solution to a precise model is “better” than a precise solution to an approximate model). To understand why, look at this situation in the following way: An approximate model usually hides the “irregularities” of a problem, thus allowing some traditional method (e. g., linear programming) to provide a precise solution. However, by hiding the irregularities of a problem, we lose much of the information needed to find the optimal solution! For example, in our simplified model above, note that the difference in transportation cost between 20 and 21 items is now the same as the difference between 19 and 20 items (which is not the case in the precise model). Consequently, the simplified model does not “see” the thresholds that play a major role in identifying the optimal solution. Thus, the “optimal” solution for an approximate model is usually more appropriate for the wastebasket than for implementation!

2.6 A Real-World Example

Let us consider a real-world example to put these business problem characteristics into context. This example is based on a “pollution control” research project that

⁵ *Linear programming* is a problem-solving method in which a linear function of a number of variables is subject to a number of constraints in the form of linear inequalities.

was completed during the late 1990s by a team of scientists from Poland and the United States. Today, we consider this project to be a forerunner of Adaptive Business Intelligence.

The main purpose of the project was to reduce ecological damage in Poland through better energy production and distribution. However, the problem was not that straightforward: The proposed solution had to reduce ecological damage *without increasing aggregate operational costs of the power stations* or *failing to meet consumer demand* (i. e., reducing the country's energy output in order to reduce ecological damage was not an option). Since the country's demand for electricity had to be satisfied, there were numerous constraints for energy production in each region. Although this problem is quite specific, it illustrates many characteristics that are common to almost all real-world business problems (e. g., an enormous number of possible solutions, many complex constraints, a time-changing environment). The similarities between this problem and other complex business problems will become apparent in the following chapters.

To solve this challenging problem, an experimental Adaptive Business Intelligence system was developed. The goal of the system was to reduce ecological damage in Poland by optimizing the energy output (and consequent pollution output) of 132 government-owned power stations. However, before we discuss the system, let us briefly address the complexity of the problem.

First, because the system was tasked with optimizing the energy production of 132 power stations, the number of possible solutions was enormous. If an integer between 1 and 10 represented the production level of each power station (i. e., 1 = 10% production, 2 = 20% production, and so forth), the number of possible solutions would be 10^{132} (10 possible solutions for one power station, 10×10 possible solutions for two power stations, $10 \times 10 \times 10$ possible solutions for three power stations, etc.). A supercomputer capable of evaluating 1,000 solutions per second would require billions of years to examine less than 1% of the search space! The fact that these power stations operated in a time-changing environment only compounded the problem, as the search space of possible solutions changed from one day to the next.

Second, the hard and soft constraints of this problem were far from trivial. Since enough energy had to be produced to supply the entire country, it was necessary to take into account the capacity and layout of the power grid in Poland. Also, although the demand for energy was constantly in flux, it was not possible to change the production levels of a particular power station too drastically. On top of everything, the system had to "protect" certain areas of Poland from long exposures to sulphur dioxide (SO_2), as some areas of the country were significantly more sensitive to SO_2 pollution than other areas.

The first step in solving this problem lay in identifying the relevant problem variables and all the relationships between them. This was accomplished by building a computational "grid model" of Poland's land area and corresponding power grid. Each square in the grid corresponded to $30 \text{ km} \times 30 \text{ km}$, which collectively covered Poland's approximate $900 \text{ km} \times 750 \text{ km}$ land area. The locations of the 132 government-owned power stations were plotted along with their energy output and emission levels, and then the pollution tolerances for each $30 \text{ km} \times 30 \text{ km}$

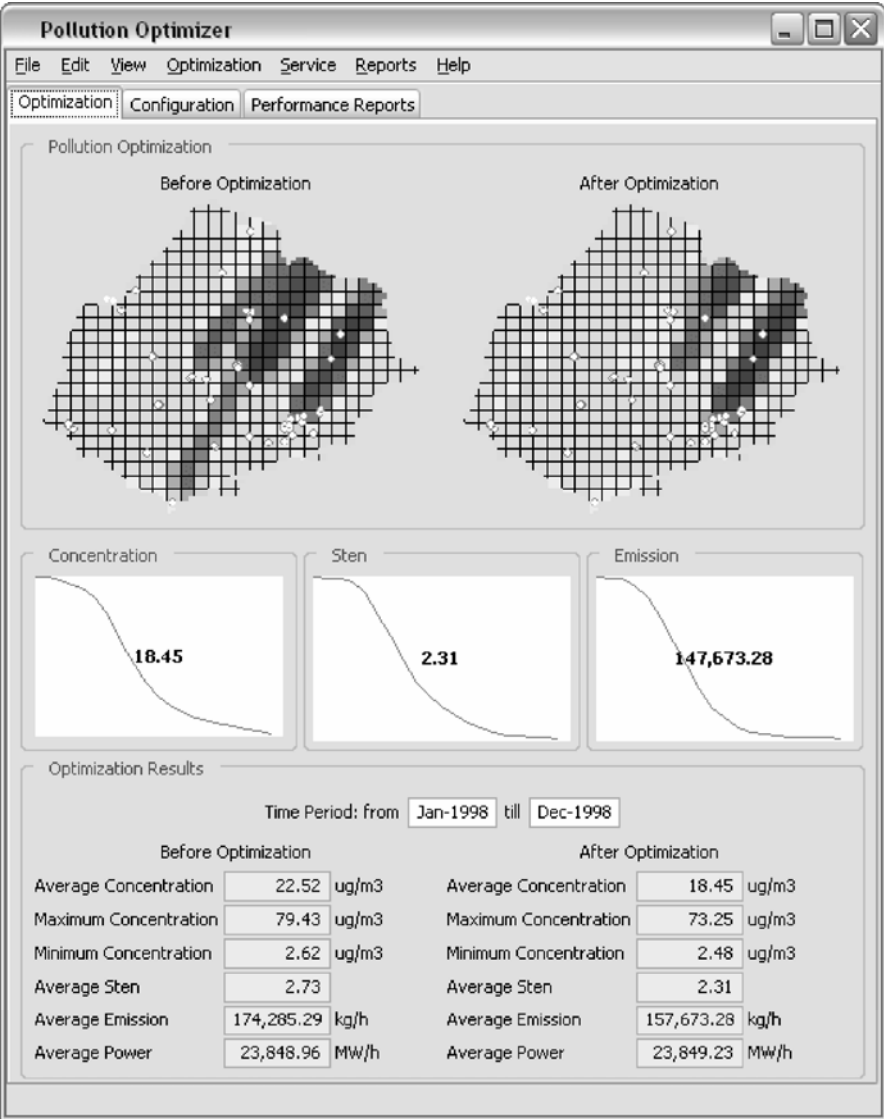
square were computed. It was also necessary to take into account the pollution caused by private enterprises in Poland, as well as many foreign sources (mainly from Germany and the Czech Republic). Because it was not possible to influence these sources, they were defined as *background concentrations*. Hence, the resulting pollution concentrations in each square were the sum of both the primary and background concentrations.

Because weather played a major role in determining how pollution affected the environment in Poland (e. g., gusty winds could easily spread high levels of pollution to remote areas of the country), it was necessary to use sophisticated weather forecasts for predicting the ecological damage that each $30 \text{ km} \times 30 \text{ km}$ square would sustain during the next 48 hours. Thus, by using data on the amount and type of pollution emitted from each power station, along with a weather forecast for the speed and direction of winds, a prediction model was developed to predict how much pollution would be created, how it would be dispersed, and how much damage it would cause in the affected squares.

To minimize the overall ecological damage in Poland (in light of the predictions made by the prediction module), an evolutionary algorithm was developed to find the optimal production level for each power station (evolutionary algorithms are discussed in Sect. 6.6). Although it was not possible to decrease the overall energy produced in Poland, it was possible to make local or regional adjustments to take advantage of various weather conditions. Furthermore, the evolutionary algorithm strived to *continuously* optimize the pollution in Poland on a daily basis. When a new weather forecast became available (along with the corresponding prediction for ecological damage), the system did not start the search from scratch; instead, it incorporated this new knowledge and continued the optimization process. Lastly, the system incorporated a module for adaptability, which compared the predicted outcome with the actual outcome. If significant prediction errors were discovered, the adaptability module would adapt the parameters of the prediction module (more on this subject in Sect. 10.3). Hence, the system was able to continuously improve its performance.

Using historical data for weather forecasts and emission levels, several experiments were conducted using this new system. The optimization process was applied to this historical data, and then the optimized results were compared with the actual, non-optimized results. From these experiments, the following conclusions were drawn: The amount of energy produced was equal in both cases, the aggregate operating costs were also equal, but the ecological damage in Poland was 15% to 18% less when the optimization process was applied. A “saving” of 15 to 18% in ecological damage is a tremendous result, especially when we consider that human lives are at risk in this particular problem. The system also contained numerous input screens, so that individual parameters could be modified to alter regional pollution tolerances (in relation to cost and service).

The sample screen below shows the optimization results. Both the “before” and “after” maps in this screen represent the country of Poland, which is divided into a grid model of $30 \text{ km} \times 30 \text{ km}$ squares (which we discussed earlier). The darker



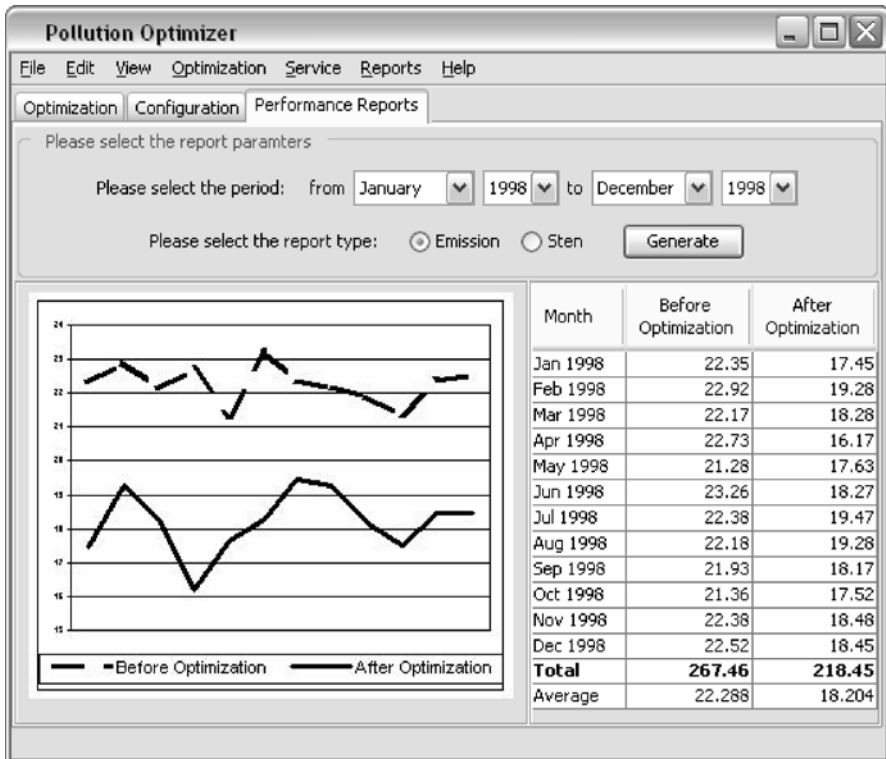
squares represent higher concentrations of pollution, and the white dots represent the 132 power stations:⁶

The left-hand-side map shows the non-optimized results for ecological damage in Poland, while the right-hand-side map shows the optimized results for the same time period. Clearly, there are fewer dark squares on the right-hand-side (optimized) map. This is reflected in the concentration and emission numbers shown at the bottom of

⁶ Some dots represent multiple power stations.

the screen. It is important to note that the system achieved these lower pollution levels while maintaining the same power production output of approximately 23,849 megawatts per hour.⁷

The report below shows the potential pollution reduction that would have been achieved in 1998:



The right-hand-side graph displays two levels of emitted toxic materials: The *Before Optimization* column displays the average monthly emissions for the default energy production scheme; and the *After Optimization* column displays the average monthly emissions for the optimized energy production scheme. The difference between the non-optimized and optimized schemes is approximately 18% ($267.46 - 218.45 = 49.01 / 267.46 = 18.32\%$), and this was accomplished without increasing the aggregate operational cost of the power stations or decreasing the amount of energy produced. By allowing the system some flexibility in these constraints, further ecological savings could be realized.

⁷ The *sten* values in the sample screen represent a scientific estimate of a region's sensitivity to pollution (mainly defined by the level of sulphur dioxide present).

As indicated at the beginning of this section, similar “savings” can be realized in other business domains, even if the particular problem is substantially different. In Chap. 4, we will discuss the overall structure of an Adaptive Business Intelligence system by outlining its key components. At that stage, it will be easy to see the similarities between this “pollution control” research project, and the core components of an Adaptive Business Intelligence system (prediction, optimization, and adaptability).

3 An Extended Example: Car Distribution

“You must drop everything, Sherlock. Never mind your usual petty puzzles of the police-court. It’s a vital international problem that you have to solve.”

The Adventure of the Bruce-Partington Plans

“I have devised seven separate explanations, each of which would cover the facts as far as we know them. But which of these is correct can only be determined by the fresh information which we shall no doubt find waiting for us.”

The Adventure of the Copper Beeches

Let us consider an extended example that most of us can relate to, which involves used cars. Specifically, let us investigate what leasing companies do with the cars that are returned at the end of a lease agreement (the so-called “off-lease” cars). We will refer to this extended example in many of the following chapters, so it is worthwhile to study this real-world business problem in detail.

3.1 Basic Terminology

However, before we discuss this example, let us first cover some basic terminology:⁸

- **Database:** A (historical) collection of data, which is the starting point for data mining and model building. Databases are usually updated on a regular basis, thereby increasing the number of stored cases. It is convenient to represent a database as a table, or a collection of tables. We can visualize the necessary data as a two-dimensional table; in this example, the table represents used cars that were sold at auction. One dimension of the table represents the number of cases (each case is a particular car), and the other dimension represents the number of variables (the characteristics of each car):

⁸ Note, however, that we will not discuss the physical organization of data (which is important from a performance perspective), but rather the data structures that support the modeling of information.

VIN	Type	Make	Model	Miles	Year	Color	Transmission	Body/Doors	Damage
2G1FP22P1P2100001	Rental	Chevy	S-10	34,983	2002	Silver	Manual	2D	\$0
WB3PF43X8X9000331	Lease	Chevy	Cavalier	59,402	2001	Red	Automatic	2D Coupe	\$0
4BBG38FJF04JDK000	Lease	Chrysler	Sebring	74,039	2000	Gray	Automatic	2D Coupe	\$500
DJOW03FFU990SJ206	Lease	Ford	Escape	37,984	2001	Green	Manual	4D Sport	\$250
JD8320DJ2094GK2X3	Rental	Ford	Focus	30,842	2001	Green	Manual	4D Sedan	\$0
2JE9F0284JD0213M3	Lease	Isuzu	Rodeo	59,044	1999	White	Automatic	4D Sport	\$250
4380JDD9W02MD001	Rental	Jeep	Cherokee	48,954	2000	Black	Automatic	4D Sport	\$500
490DK20285JF0209D	Rental	Mazda	626	38,943	2000	White	Automatic	4D Sedan	\$0
10D92JD920KD00002	Lease	Nissan	Altima	39,488	2000	Black	Automatic	4D Sedan	\$0
D920DKJ0284JJ9990	Rental	Nissan	Altima	23,584	1999	White	Manual	4D Sedan	\$0
JD88D92JJD02K3361	Rental	Saturn	L	21,048	2001	White	Automatic	4D Sedan	\$750
10DS0JJ20DXI00093	Lease	Suzuki	Vitara	15,849	2003	Yellow	Automatic	2D Sport	\$0
21KD02KD0DJ920M27	Lease	BMW	Z3	49,858	2000	Blue	Manual	2.3 RSTR	\$250
389DJ2DD298JWQ082	Lease	Ford	Explorer	42,893	2002	Green	Automatic	XLT 4WD	\$0
108DJ2048FJJ20043	Rental	Ford	Mustang	20,384	2002	Red	Manual	GT	\$0
DJC82002009DD2J04	Rental	Mercury	Frontier	27,849	2001	Silver	Automatic	SE-V6 Crew	\$500
830DMM3029XMM0092	Lease	Honda	Accord	26,849	2002	Yellow	Automatic	EX V6	\$0
CNEU200220CCI2202	Rental	Toyota	4Runner	33,483	2000	Silver	Automatic	SR5	\$0
CNDJ2940JD88D2JD0	Lease	VW	Beetle	5,459	2003	Blue	Manual	GLS 1.8T	\$0
1VC0CMEJ200V9EJJ1	Lease	Toyota	4Runner	81,837	2001	Silver	Automatic	LMTD 4WD	\$250

- **Case:** A data structure that represents a single occurrence of recorded phenomenon (e. g., a sale, transaction). Referring to our database of sold cars, each case provides a description of a car (VIN,⁹ make, model, mileage, etc.) together with transaction details (sale price, date, location). Cases (also called *records*, *instances*, and *examples*) correspond to the horizontal rows in database tables.
- **Variable:** The smallest data unit in a case. Variables (also called *attributes*, *fields*, *features*, *dimensions*, and *characteristics*) correspond to the vertical columns in a table. For example, in our database of sold cars, there are many variables (e. g., “color,” “mileage”), and each variable takes its value from a predefined set. For example, the variable “color” can take any of the following values: “white,” “silver,” “beige,” etc. The variable “mileage” (marked in the above table as “miles”) can take any integer as a value (e. g., 34,789).
- **Prediction method:** A particular method that is used to build a prediction model from a data set. For example, we can use the *linear regression* method to develop a linear regression model (covered in Sect. 5.2.1).
- **Prediction model:** The output of a prediction method. Different prediction models have different characteristics, advantages, and disadvantages (e. g., response time, precision, ease of updating the model).
- **Parameter:** A number that defines the relative “merit” of a variable that is used in a prediction model. Every model has a few parameters (sometimes called *coefficients* or *weights*). For example, a linear regression model for predicting the sale price of a particular car at a particular location may be defined by a function:

$$\text{Sale Price} = a + (b \times \text{Mileage}) + (c \times \text{Year}) + (d \times \text{Color}) + \dots$$

which provides the predicted price for a new case when supplied with the numeric values of the other variables (“mileage,” “year,” “color,” etc.). The numbers a , b , c , d , ... are the parameters of the model, and they define how much “weight” the model gives to each variable (in this example, parameter a is a constant). For example, if the parameters b and c are set to -0.6 and -0.2 in the above function, then this means that the model will put three times more weight on the variable “mileage” than on the variable “year” when predicting the sale price. The parameters of a model require careful tuning during the training phase.

Because everything begins with data, these definitions are a prerequisite to our discussions. The other relevant terms will be provided in the course of the text.

3.2 Off-lease Cars

When a lease agreement expires, the off-lease car is either returned to the leasing company or purchased by the lessee. The leasing company does not worry about the cars that are purchased, but the cars that are returned have to be sold at auction.

⁹ VIN is an acronym for “vehicle identification number,” which is a string of 17 digits and letters that contains considerable information about a specific vehicle, (including country of origin, manufacturer, and model year).

Each returned car is different in its make, model, body style, trim, color, year, mileage, and damage level.

A leasing company maintains this information in an inventory database, as the one displayed in the previous section. Because a car’s characteristics will strongly influence its sale price, leasing companies also keep careful track of this information. The sales data are typically kept in a database that contains the VIN, ZIP code of the auction site (there are hundreds of different auction sites in the United States), date, and the sale price of each car:¹⁰

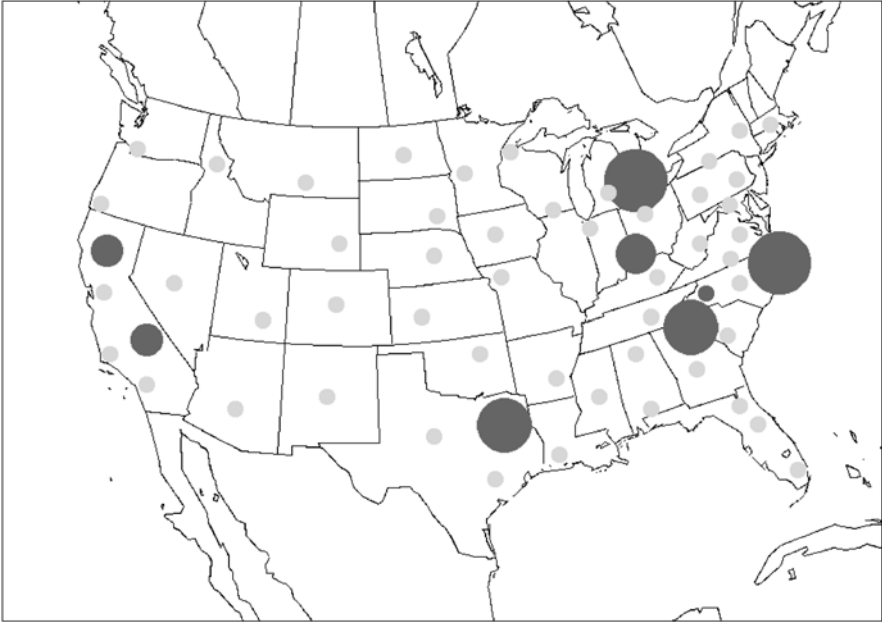
VIN	ZIP	Date	Price
39WWK93309KJ33012	28262	2.11.2004	\$12,035
UDJ2293M99DL0K220	30334	2.11.2004	\$15,600
4D09WJD92JE93H990	30334	2.11.2004	\$10,590
KD37D92JF83NF8822	90012	3.11.2004	\$9,265
NKI2389DD974F2235	28262	3.11.2004	\$13,450
K29DH38FHW02HD923	48243	3.11.2004	\$13,955
MDK293HFDWH299305	90012	4.11.2004	\$12,495
28DN39FNDJW2N0024	90012	4.11.2004	\$11,925
29H93NFI3HJF93F04	48243	4.11.2004	\$11,396
ND920ENF1NAD02834	48243	5.11.2004	\$9,835
D39DJ39EHQ8HH9335	28262	5.11.2004	\$8,965
02UFIMF03JF9SH935	90012	5.11.2004	\$13,960
D932NF93HG9057362	48243	5.11.2004	\$8,830
00F8EB3IDNB293758	48243	8.11.2004	\$7,920
IE038THJ203TH0234	28262	8.11.2004	\$19,250
39FH324MV092HGM39	48243	8.11.2004	\$22,640
F92N9F389FH120458	90012	8.11.2004	\$13,580
F9485JG03H25495J5	30334	9.11.2004	\$16,970
08GN94HJH03J49327	30334	9.11.2004	\$14,320
F04JH402KG4509G45	48243	9.11.2004	\$9,110

Given all of this information, the question is: Where should leasing companies send their off-lease cars to get the best price? This is a difficult question to answer, especially when we consider that some leasing companies lease more than *one million* cars each year. That is more than 3,000 cars per day – a truly staggering figure!

3.3 The Problem

Now, imagine that we are in charge of selling off-lease cars for a fictitious leasing company. Let us assume that the company receives 3,000 off-lease cars each day and does business with 50 designated auction sites. The following figure illustrates the case (for a particular day of operation):

¹⁰ We could easily obtain the characteristics of each car by merging it with the previous table.



The darker circles represent the locations of the returned cars; the larger the circle, the more cars were returned in that location (clearly, most of the cars were returned on the east coast of the United States). Note that the sizes and locations of the darker circles will vary from one day to the next, as different people and organizations will return their cars at different locations. The lighter circles of the same size, on the other hand, represent the 50 auction sites that are available for selling the returned off-lease cars. The locations of these auction sites are fixed.¹¹

Our task is to distribute a daily load of 3,000 cars to these 50 auction sites. In other words, we have to assign an auction site to each car. For example, the first car may be located at a dealership in Northern California, and we have to make a decision where to ship it. At first glance, this looks easy. We might be tempted to take one car at a time, consult some report¹² on what the average sale price for that particular car is at each auction site (after adjusting for mileage, trim, damage level, etc.), and then ship the car to the auction site with the highest average sale

¹¹ Although the locations of the 50 auction sites are fixed, the company may, from time to time, change the auctions it does business with by dropping some sites and adding new ones (thereby changing the layout of the 50 lighter circles). This may happen if cars are routinely damaged at some sites, auction fees go up, or some other reason. However, these decisions raise several additional questions, such as: *How do we evaluate the monetary impact of dropping some sites and adding others?* and, *Can we increase profits by replacing some auction sites with others?* We will address these important questions later in the text.

¹² Many reports are available for estimating the auction price of cars, including *Black Book*, *Kelley Blue Book*, the *Manheim Market Report*, and others.

price. Of course, we would also have to estimate the transportation cost (the longer the distance, the higher the cost) to each auction site, but these calculations are quite manageable. Using this method, our decision for the first car can be visualized in the following way:

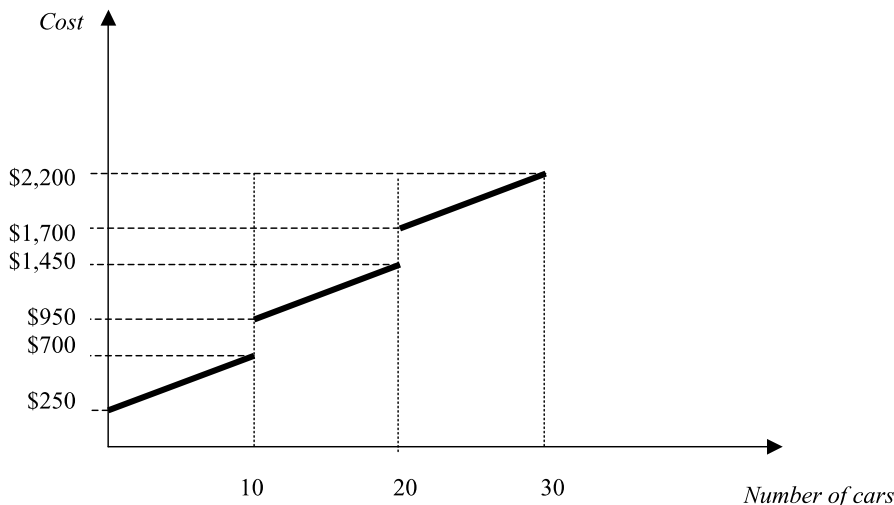


The line represents our decision to ship the car from Northern California (larger circle) to an auction site in Idaho (smaller circle). We would have to repeat this decision process for each car, and, although tedious, it is definitely doable. To minimize the amount of work involved, we can also write a simple program to do this tedious work for us.

So, what is the problem? Well, simply put, the problem is that the above approach does not work very well. To see why, we have to delve a little deeper into the problem.

3.4 Transportation

When we ship an entire truckload of cars from one place to another, we get a cheaper rate per car than if we send only one car (or a few cars) at a time. This occurs because the major component of the transportation cost is the truck, and it is of lesser importance how many cars are actually on the truck. Hence, the relationship between the transportation cost (between two fixed locations) and the number of transported cars may look something like this model (recall the discussion on transportation cost models from the previous chapter):



Given the above model, the cost for sending a single car from one location to another is \$250, but the cost of sending two cars is “only” \$300 (note that the cost per car is then \$150). Then there are \$50 increments per car up to 10 cars (as we can load all of them onto a single truck). Thus, shipping 10 cars would cost \$700, or just \$70 per car. Note that the increments from one car to the next for the first 10 cars need not be linear. For example, the truck might have two levels and loading a car onto the upper level might be slightly more expensive than loading a car onto the lower level. For this reason, a straight line need not represent the “1 to 10 cars” part of the graph. Furthermore, if we want to transport 11 cars, then we will experience a “jump” in cost: We will pay \$700 for 10 cars on the first truck, plus \$250 for a single car on the second truck (for a total of \$950).

Because most transportation companies have trucks of different shapes and sizes (which can hold different amounts of cars), a more realistic transportation cost model may look something like (again, we assume fixed locations for “to” and “from”):

- 1–6 cars: \$120 per car.
- 7–10 cars: \$95 per car.
- 11–14 cars: \$85 per car.
- The cost of transporting more than 14 cars is calculated as follows: \$85 per car for multiples of 14 cars (such as 28, 42, 56, etc.), \$120 per car for the remaining 1–6 cars, \$95 per car for the remaining 7–10 cars, and \$120 per car for the remaining 10–13 cars. Transporting 20 cars, for example, would cost \$85 per car for the first 14 cars and then \$120 per car for the remaining 6 cars, for a total of \$1,910 ($14 \times \$85 + 6 \times \120).

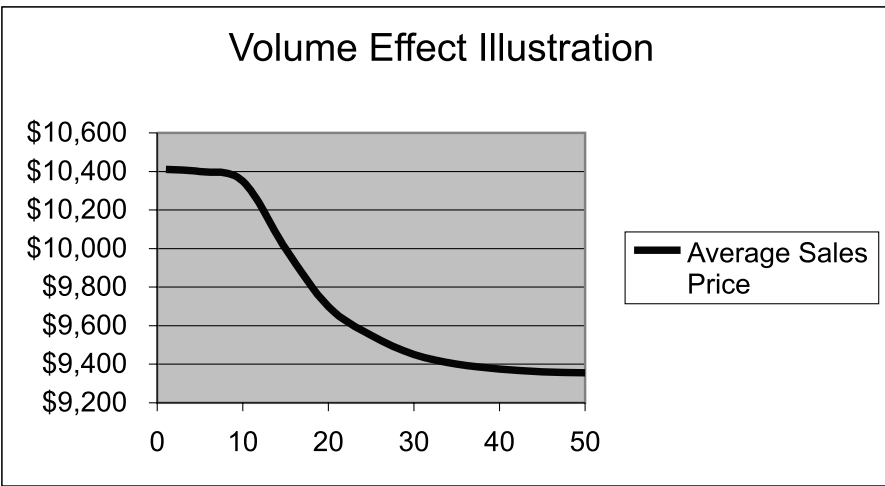
In addition to transportation cost, we also have to deal with numerous other issues, such as depreciation, auction calendars (auctions are not organized every

day), and risk (cars can fall down from trucks and get damaged, cars might get stolen, etc.). We will discuss some of these later in this chapter.

3.5 Volume Effect

Although our goal is to ship each car to the auction site that offers the best price, we might inadvertently trigger the “volume effect” by sending too many cars of same color, make, and mileage to the same auction site. To understand the volume effect, imagine an auction site that holds car auctions every second Wednesday at 10 am. Used-car buyers come to this auction site in the morning to inspect the cars, choose the ones they want, and decide how much to bid. Now imagine what would happen if we sent 45 white Ford Mustangs to this auction site: In all probability, these cars would sell for the minimum opening price. With 45 identical cars for sale, most buyers would be reluctant to bid up the price. On the other hand, imagine if we sent only one white Ford Mustang to this auction site: There would be several interested buyers, and they would bid up the price of the car. This volume effect is extremely important, as the higher the auction sale price, the more money the company will make.

The volume effect for a particular car at a particular auction site might look something like:



This graph illustrates the volume effect phenomenon: We get more money per car by selling *fewer* similar cars. For example, the current average sale price for a particular car at a particular auction site might be \$10,400. We can get the same price if we ship up to seven cars to this location. However, if we ship 30 similar cars, then the average sale price per car will drop to \$9,450. Note that the term “similar” can mean more than just the same make/model or color. For example,

many white compact cars of different makes and models would still compete for the same buyers, thereby reducing the average sale price per car.

Consequently, we cannot consider one car at a time – we have to consider the collective effect of all the cars. This means that for one car we have 50 possible solutions (as we can ship this car to one of 50 auction sites), for two cars we have 2,500 possible solutions (50×50), for three cars we have 125,000 possible solutions ($50 \times 50 \times 50$), etc. For 3,000 cars, we have approximately 50^{3000} possible solutions (50 multiplied by itself 3,000 times)! This is an overwhelming number (larger than the estimated number of atoms in the Universe) and no supercomputer can evaluate all the possible combinations in a billion human lifetimes. Nevertheless, we have to make a decision for all of these cars *today*!

3.6 Price Depreciation and Inventory

To further complicate matters, every auction site has a set day for selling cars (e. g., every second Friday at 10 am). Because of this, if we ship 100 cars to an auction site and the delivery arrives one or two days *after* the auction day, then the cars will have to wait until the next sale day. This is bad for the leasing company, because on top of other reasons (such as paying the auction sites to keep the cars on their premises for up to two weeks), the cars will also lose value on a daily basis. This phenomenon is called “price depreciation,” and it amounts to approximately \$10 per day, per average car.

Also, we might already have 1,000 cars at a particular auction site, but only 250 of our cars are sold (on average) at each sale. Hence, it will take approximately four auction sales to sell our current inventory. This means that if we ship an additional 100 cars to this particular auction, they would be sold some two months later. Therefore, we have to take into account our current inventory at each auction site, as it will affect the price depreciation on future shipments. Of course, our inventory should also include cars that are in transit to the auction, and before we can make any distribution decision, it would be necessary to check our inventory levels at *all* 50 auction sites. Otherwise, it would be next to impossible to make a smart decision on where to ship today’s load of cars (this is especially important for large leasing companies, which may have 50,000 or more cars sitting in inventory).

3.7 Dynamic Market Changes

Another issue that makes our job more difficult is the fluctuation in used car prices. These fluctuations may be slow and subtle, or sudden and dramatic (as was the case after the September 11th terrorist attacks), and they are often region specific (e. g., it is better to sell convertibles in sunny Florida during the wintertime than in snowy Boston – something called the “seasonality effect”). These changing market conditions force us to stay on top of price changes at each auction site – something that is very difficult to do. We also have to deal with next year’s

models entering the marketplace during the months of August and September, as the older models will immediately drop in price when this happens (also part of the “seasonality effect”). This is an important consideration when we approach this time of year, as it might be better to ship cars nearby and sell them “in time,” rather than sending them far away to fetch a better price. Additionally, new body style models are introduced every few years, which cause an even bigger drop in price for the older body styles.

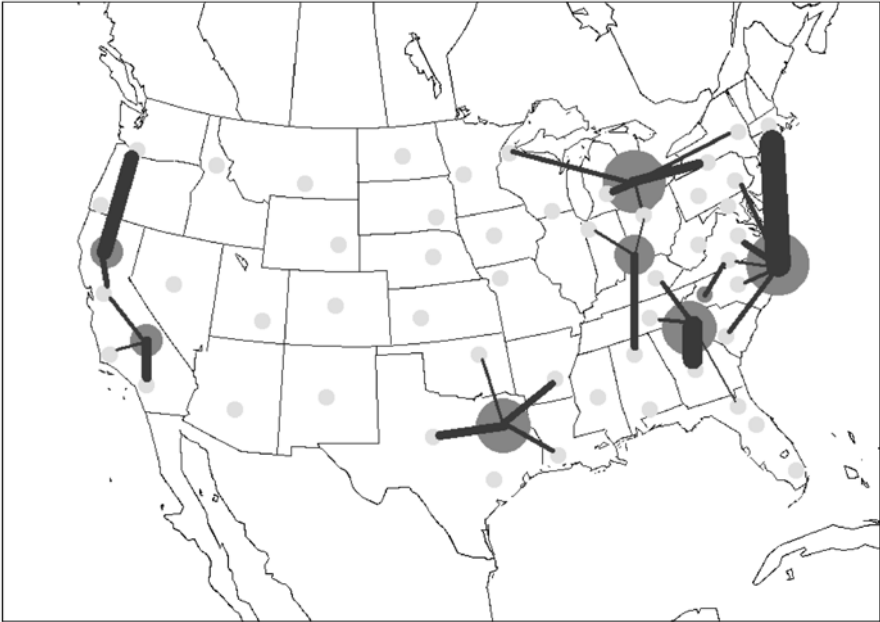
And that is not all. Note that it takes some time to transport a car to a specific auction site. The truck has to drive to the pick-up location, load the car, pick up some additional cars (possibly somewhere close by), and then, finally, drive the cars to the assigned auction site. This process can take two weeks (or longer for shipments from the East Coast to the West Coast), and during this time the prices might change at the auction site. Hence, we have to predict the sale price for each car a couple of weeks ahead of time, and take into account seasonality and other changes in the market. Again, let us illustrate this by an example.

Say we are considering several possible auction sites for a car located in Jacksonville, Florida. Specifically, we are considering sending the car to an auction site in Georgia, Pennsylvania, or California. The price estimations for these three auction sites are different, because we are interested in an estimated sale price for 5 days from today for the Georgia auction site, an estimated sale price for 10 days from today for the Pennsylvania auction site, and an estimated sale price for 15 days from today for the California auction site. The differences in time are due to the transportation distance. However, to estimate these prices, we have to take into account factors such as the seasonality effect and price depreciation, and these additional factors make our decision a bit tricky.

In making the decision of Georgia vs. Pennsylvania vs. California, we also have to take into account transportation costs, the volume effect, and current inventories (as discussed earlier). We also have to weigh the possibility of a better price in California against higher transportation costs, higher price depreciation, higher risk, etc. The decision is not an easy one.

3.8 The Solution

Without a doubt, this is not a trivial problem. In order to make a decision, we have to consider the characteristics of each car, many different possible auction sites, complex transportation costs, volume effects, countrywide inventory of cars (or cars in transit to the auction), price depreciation, and market-driven changes in price. At the end of the day, we do not know whether our decisions are “optimal” – whether they maximize the potential profit for the company, or whether there is a better solution – but we can visualize the overall distribution for a particular day as follows:



In the figure above, the lines represent the transportation connections between the car locations and auction sites, and the thicknesses of these lines represent the transported volume (the thicker the line, the more cars are shipped). Because the problem is so complex, every leasing company has a dedicated team for assigning off-lease cars to auctions sites. These teams face the formidable task of recommending the best possible solution for each daily load of cars. A small mistake, an inferior recommendation that results in a net loss of “only” \$150 per car, may cost the company hundreds of thousands of dollars in a single day!

On the other hand, if an Adaptive Business Intelligence system was used to improve the daily distribution of cars to the tune of \$200 per car (note here that this corresponds to an increase of only 1.33% in the price of an average car), then the increased annual profits might translate into hundreds of millions of dollars! Hence, to maximize the overall net profit, an Adaptive Business Intelligence system must *decide* where to ship each car. As discussed earlier, the problem is extremely complex, and the system would have to address the following issues:

- *Transportation.* When a whole truckload of cars is shipped from one place to another, the company is charged a cheaper rate. Therefore, it is important to take into account the number of cars that are transported on each truck.
- *Risk factors.* Cars can be damaged, stolen, or the transportation truck might be involved in an accident. Longer trips also increase the probability of a delay.
- *Volume effect.* If many cars of the same type are sent to the same auction site, then the volume effect will kick in and an oversupply of a particular car will decrease its price.

- *Size of the search space.* The distribution of 3,000 cars to 50 auction sites gives us 50^{3000} possible solutions, which is much larger than the estimated number of atoms in the Universe.
- *Price depreciation.* Every auction site has a typical sale day (e. g., every Wednesday at 10 am, or every second Thursday at 11 am). If some cars arrive one day after the sale date, then they will sit at the auction site for one or two weeks (until the next sale day), and the price depreciation is often around \$10 per day, per average car.
- *Recent history.* When making a recommendation, all decisions made during the past few weeks must be taken into account. Many of those cars may still be in transit, and if they are going to the same auction site, then they might be sold on the same day.
- *Inventory.* It is important to monitor the inventory level of cars at all auction sites, as each site has a particular throughput. If an auction can handle 250 cars per sale day, and the current inventory is larger than 250, then additional time must be added to the estimated sale date.
- *Dynamic market conditions.* Market prices for cars change quite frequently, sometimes slowly and sometimes very quickly. Leasing companies (like most businesses) operate in a non-stationary environment that is influenced by many external factors, such as: (1) seasonality (e. g., it is not easy to sell convertibles in New York during the wintertime), (2) the arrival of new models (e. g., new models enter the marketplace in August, influencing the price of older models), and (3) weather (which influences the number of dealers present at an auction, which in turn influences the sale price).
- *Business rules.* It is essential to accommodate various business rules that can be added or dropped at any time (e. g., “do not send any red cars to South-East auctions”). This is important for analyzing what-if scenarios.

Let us conclude this chapter with an observation that we have already discussed (in general terms) in Chap. 1: namely, *that all the knowledge in the world will not guarantee the right or best decision*. Hence, even if we possessed “perfect knowledge” and were capable of accurately predicting the price of *any* car at *any* auction site for *any* day, we still would not know how to optimally distribute 3,000 cars on any given day because of all the issues mentioned above. The number of possible distributions is simply too large to be processed in any reasonable amount of time. Hence, even though it is true that data mining can extract useful knowledge from data, it is a myth that the extraction of such knowledge will make a business run better! We may have perfect knowledge of all cars, prices, and locations, and still our distribution decisions might be very bad!

4 Adaptive Business Intelligence

“By Jove, Mr. Holmes, I think you have hit it.”
The Adventure of the Lion’s Mane

“ ‘Elementary,’ said he.”
The Adventure of the Crooked Man

In Chap. 2, we discussed the characteristics of real-world business problems, which included:

- The number of possible solutions to the problem is so large that it prohibits any systematic (complete) search for the best solution.
- The problem exists in a time-changing environment, and therefore requires a set of solutions (rather than a single solution).
- The problem is so heavily constrained that finding even one feasible solution is often difficult, let alone searching for the optimum solution (or set of solutions).
- The problem includes many (possibly conflicting) objectives.

In the previous chapter, we gave an example of a real-world business problem in the car leasing industry. There is no question that the number of possible solutions to the car distribution problem is too large for any systematic search (we indicated that the distribution of 3,000 cars to 50 auction sites gives us 50^{3000} possible solutions). We may also want to impose many constraints. For example, we may wish to restrict the maximum transportation distance (e. g., “cars should not be shipped more than 1,000 miles from their original location”), exclude some types of cars from some auction sites (e. g., “red cars should not be sent to auction sites in Texas” or “cars with more than 100,000 miles should not be sent to auctions in the South-East”), etc. Note also, that in our example:

- The number of returned cars changes every day.
- The locations where the cars are returned changes every day.
- The prices at each auction site change over time.
- The number of available auction sites may change over time.
- The volume effect changes over time.
- Transportation costs change over time, etc.

Because all organizations operate in a time-changing environment, they are forced to constantly adapt and adjust. Consequently, an Adaptive Business Intelligence system must include three major components: (1) a component for making predictions (in our case, sale price predictions), (2) a component for making near-optimal decisions (in our case, the distribution of cars), and (3) a component for adapting the prediction module to changes in the environment. To create such a system, the following four steps should be followed:¹³

- The available data must be prepared and thoroughly analyzed (the data mining stage).
- A prediction model must be developed based on the data mining results.
- An optimization module must be developed that uses the prediction model to recommend the best solution.
- An adaptability module must be developed that is responsible for adapting the prediction module to the time-changing environment.

Let us discuss the details of these seemingly simple steps, as understanding them will enable us to realize the powerful benefits of Adaptive Business Intelligence!

4.1 Data Mining

Data typically reside in one or more databases, and we first have to understand the structure of each database and its tables. To illustrate this further, let us use a table from Chap. 3 that has basic data about cars and their sale prices:

¹³ Before you can solve a problem, you have to understand it. In software development, this means gaining a complete understanding of the user requirements. Unfortunately, this process can be quite tricky, because business managers understand business processes while software developers understand software development. To bridge this gap, software developers should talk directly with business managers to understand the relevant business process and user requirements. With a complete understanding of the problem, software developers can then create a user requirements document that is verified by the business managers. Because software development is covered many specialized texts, we will instead focus on the conceptual issues involved in developing an Adaptive Business Intelligence system.

VIN	ZIP	Date	Price
39WWK93309KJ33012	28262	2.11.2004	\$12,035
UDJ2293M99DL0K220	30334	2.11.2004	\$15,600
4D09WJD92JE93H990	30334	2.11.2004	\$10,590
KD37D92JF83NF8822	90012	3.11.2004	\$9,265
NKI2389DD974F2235	28262	3.11.2004	\$13,450
K29DH38FHW02HD923	48243	3.11.2004	\$13,955
MDK293HFDWH299305	90012	4.11.2004	\$12,495
28DN39FNDJW2N0024	90012	4.11.2004	\$11,925
29H93NFI3HJF93F04	48243	4.11.2004	\$11,396
ND920ENF1NAD02834	48243	5.11.2004	\$9,835
D39DJ39EHQ8HH9335	28262	5.11.2004	\$8,965
02UFIMF03JF9SH935	90012	5.11.2004	\$13,960
D932NF93HG9057362	48243	5.11.2004	\$8,830
00F8EB3IDNB293758	48243	8.11.2004	\$7,920
IE038THJ203TH0234	28262	8.11.2004	\$19,250
39FH324MV092HGM39	48243	8.11.2004	\$22,640
F92N9F389FH120458	90012	8.11.2004	\$13,580
F9485JG03H25495J5	30334	9.11.2004	\$16,970
08GN94HJH03J49327	30334	9.11.2004	\$14,320
F04JH402KG4509G45	48243	9.11.2004	\$9,110

And another table that has more detailed data about each car:

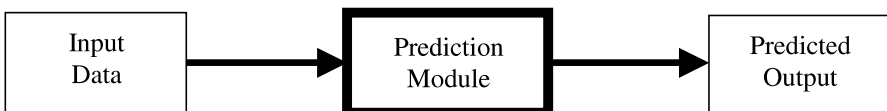
VIN	Type	Make	Model	Miles	Year	Color	Transmission	Body/Doors	Damage
2G1FP22P1P2100001	Rental	Chevy	S-10	34,983	2002	Silver	Manual	2D	\$0
WB3PF43X8X9000331	Lease	Chevy	Cavalier	59,402	2001	Red	Automatic	2D Coupe	\$0
4BBG38FJF04JDK000	Lease	Chrysler	Sebring	74,039	2000	Gray	Automatic	2D Coupe	\$500
DJOW03FFU990SJ206	Lease	Ford	Escape	37,984	2001	Green	Manual	4D Sport	\$250
JD8320DJ2094GK2X3	Rental	Ford	Focus	30,842	2001	Green	Manual	4D Sedan	\$0
2JE9F0284JD0213M3	Lease	Isuzu	Rodeo	59,044	1999	White	Automatic	4D Sport	\$250
4380JDD9W02MD001	Rental	Jeep	Cherokee	48,954	2000	Black	Automatic	4D Sport	\$500
490DK20285JF0209D	Rental	Mazda	626	38,943	2000	White	Automatic	4D Sedan	\$0
10D92JD920KD00002	Lease	Nissan	Altima	39,488	2000	Black	Automatic	4D Sedan	\$0
D920DKJ0284JJ9990	Rental	Nissan	Altima	23,584	1999	White	Manual	4D Sedan	\$0
JD88D92JJD02K3361	Rental	Saturn	L	21,048	2001	White	Automatic	4D Sedan	\$750
10DS0JJ20DXI00093	Lease	Suzuki	Vitara	15,849	2003	Yellow	Automatic	2D Sport	\$0
21KD02KD0DJ920M27	Lease	BMW	Z3	49,858	2000	Blue	Manual	2.3 RSTR	\$250
389DJ2DD298JWQ082	Lease	Ford	Explorer	42,893	2002	Green	Automatic	XL/T 4WD	\$0
108DJ2048FJJ20043	Rental	Ford	Mustang	20,384	2002	Red	Manual	GT	\$0
DJC82002009DD2J04	Rental	Mercury	Frontier	27,849	2001	Silver	Automatic	SE-V6 Crew	\$500
830DMM3029XMM0092	Lease	Honda	Accord	26,849	2002	Yellow	Automatic	EX V6	\$0
CNEU200220CCI2202	Rental	Toyota	4Runner	33,483	2000	Silver	Automatic	SR5	\$0
CNDJ2940JD88D2JD0	Lease	VW	Beetle	5,459	2003	Blue	Manual	GLS 1.8T	\$0
1VC0CMEJ200V9EJJ1	Lease	Toyota	4Runner	81,837	2001	Silver	Automatic	LMTD 4WD	\$250

Since we have explained what each column means in the previous chapter,¹⁴ we should already have a good understanding of the data. Next, we have to clean the data. “Clean the data?” Yes, data can be “dirty”! This terminology refers to situations that involve missing data, inconsistent data, incorrect data, etc. Manually entered data usually has to be cleaned since humans make typing errors and often use different words to describe the same thing (e. g., “Chevrolet” and “Chevy”). Cleaning the data means identifying what data might be missing, what data might have been manually entered, what data might be inconsistent or incorrect, and so on (we will discuss the process of preparing and cleaning data in more detail in the next chapter).

In this small sample, there are no missing values or irregularities, so we do not have to worry about dirty data. We can now proceed with the *data mining* activity, which is an analytical process for finding relationships and patterns among variables (e. g., finding the relationship between “sale price,” “make,” “model,” and “mileage”). Although *knowledge discovery* (e. g., text mining, discovery of association rules) is oftentimes an important goal of data mining, we are more interested in using the data mining results to build a model (or set of models) for predicting some response (e. g., the outcome of a loan application, the probability that a credit card transaction is fraudulent). The reason for this is that predictions are directly applicable to decision making, whereas knowledge discovery is closer to decision support. In the car distribution example, we are interested in predicting the sale price for particular off-lease cars at particular auction sites at particular dates in the future. Therefore, we should focus our data mining efforts on finding the relationships and patterns that exist among the various variables in the data set, and then use this knowledge to develop an effective model for predicting the sale price.

4.2 Prediction

The basic function of a prediction module (which can made up of one or more prediction models) is to produce an output based on some input:



To make the prediction module functional, it is necessary to “train” the various underlying models using historical data. During this process, the prediction module “learns” how to predict the output given the input (we explain this process in detail in Chap. 5). As an example, imagine predicting what a particular Honda Accord (with known features and known mileage) would sell for at a particular auction site in Northern California at a particular point in time (e. g., in two weeks). In this case, the input would be the auction site in Northern California, the

¹⁴ In many data mining activities, understanding what the columns mean and how they are related is not an easy task!